

## SAFETY DATA SHEET

Creation Date 13-May-2009

Revision Date 28-March-2024

Revision Number 3

### 1. Identification

**Product Name** D-Serine methyl ester hydrochloride

**Cat No. :** H63685

**CAS-No** 5874-57-7

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Serious Eye Damage/Eye Irritation**

Category 1

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Causes serious eye damage

**Precautionary Statements****Prevention**

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                             | CAS-No    | Weight % |
|---------------------------------------|-----------|----------|
| D-Serine, methyl ester, hydrochloride | 5874-57-7 | >95      |

### 4. First-aid measures

|                                                               |                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                                         | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                                             | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | Causes severe eye damage.<br>Treat symptomatically                                                                |

### 5. Fire-fighting measures

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                   |
| <b>Flash Point</b>                      | Not applicable                             |
| <b>Method -</b>                         | No information available                   |
| <b>Autoignition Temperature</b>         | No information available                   |
| <b>Explosion Limits</b>                 |                                            |
| <b>Upper</b>                            | No data available                          |
| <b>Lower</b>                            | No data available                          |
| <b>Sensitivity to Mechanical Impact</b> | No information available                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                   |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

**7. Handling and storage****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

**Storage.**

Keep container tightly closed in a dry and well-ventilated place.  
To maintain product quality. Keep refrigerated. Protect from moisture. Incompatible Materials. Strong oxidizing agents. Strong acids.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.  
Ensure adequate ventilation, especially in confined areas.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Glove material**

Nitrile rubber  
Neoprene  
Natural rubber  
PVC

**Breakthrough time**

See manufacturers  
recommendations

**Glove thickness**

-

**Glove comments**

Splash protection only

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                                 |
|-----------------------------------------------|---------------------------------|
| <b>Physical State</b>                         | Solid                           |
| <b>Appearance</b>                             | White                           |
| <b>Odor</b>                                   | No information available        |
| <b>Odor Threshold</b>                         | No information available        |
| <b>pH</b>                                     | No information available        |
| <b>Melting Point/Range</b>                    | 152 - 161 °C / 305.6 - 321.8 °F |
| <b>Boiling Point/Range</b>                    | No information available        |
| <b>Flash Point</b>                            | Not applicable                  |
| <b>Evaporation Rate</b>                       | Not applicable                  |
| <b>Flammability (solid,gas)</b>               | No information available        |
| <b>Flammability or explosive limits</b>       |                                 |
| Upper                                         | No data available               |
| Lower                                         | No data available               |
| <b>Vapor Pressure</b>                         | No information available        |
| <b>Vapor Density</b>                          | Not applicable                  |
| <b>Specific Gravity</b>                       | No information available        |
| <b>Solubility</b>                             | Soluble                         |
| <b>Partition coefficient; n-octanol/water</b> | No data available               |
| <b>Autoignition Temperature</b>               | No information available        |
| <b>Decomposition Temperature</b>              | No information available        |
| <b>Viscosity</b>                              | Not applicable                  |
| <b>Molecular Formula</b>                      | C4 H9 N O3 . Cl H               |
| <b>Molecular Weight</b>                       | 155.58                          |

## 10. Stability and reactivity

|                                         |                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                                         |
| <b>Stability</b>                        | Stable under recommended storage conditions. Hygroscopic.                                                          |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation. Exposure to moist air or water.                          |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids                                                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Hydrogen chloride gas |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                          |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                      |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Mist LC50

Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.

#### Component Information

##### Toxicologically Synergistic Products

No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

#### Irritation

No information available

#### Sensitization

No information available

#### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                             | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------|-----------|------------|------------|------------|------------|------------|
| D-Serine, methyl ester, hydrochloride | 5874-57-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

#### Mutagenic Effects

No information available

#### Reproductive Effects

No information available.

#### Developmental Effects

No information available.

#### Teratogenicity

No information available.

#### STOT - single exposure

None known

#### STOT - repeated exposure

None known

#### Aspiration hazard

No information available

#### Symptoms / effects, both acute and delayed

No information available

#### Endocrine Disruptor Information

No information available

#### Other Adverse Effects

The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

#### Persistence and Degradability

Soluble in water Persistence is unlikely based on information available.

#### Bioaccumulation/ Accumulation

No information available.

#### Mobility

Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

### Waste Disposal Methods

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

Not regulated

|                 |               |
|-----------------|---------------|
| <u>TDG</u>      | Not regulated |
| <u>IATA</u>     | Not regulated |
| <u>IMDG/IMO</u> | Not regulated |

## 15. Regulatory information

### International Inventories

| Component                             | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|---------------------------------------|-----------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| D-Serine, methyl ester, hydrochloride | 5874-57-7 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                             | CAS-No    | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------------------|-----------|-------|------|------|------|------|------|-------|-------|
| D-Serine, methyl ester, hydrochloride | 5874-57-7 | -     | -    | -    | X    | -    | -    | X     | -     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                             | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| D-Serine, methyl ester, hydrochloride | 5874-57-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                             | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| D-Serine, methyl ester, hydrochloride | 5874-57-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By Product Safety Department  
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[www.thermofisher.com](http://www.thermofisher.com)

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Creation Date</b>    | 13-May-2009                                        |
| <b>Revision Date</b>    | 28-March-2024                                      |
| <b>Print Date</b>       | 28-March-2024                                      |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**Disclaimer**

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**End of SDS**